INDIVIDUAL ASSIGNMENT1

Integrated Situation: D&G E-Commerce Mobile Application Using React Native

D&G is a growing online retail start-up located in Rwamagana, Rwanda. The company aims to strengthen its presence in the digital marketplace by providing customers with a seamless and user-friendly shopping experience. It offers a wide range of products, including clothes, shoes, bags, household equipment, jewellery, and cosmetics.

To achieve its business objectives, D&G is hiring a React Native mobile application developer to design and implement a comprehensive cross-platform e-commerce mobile application that runs on both Android and iOS devices. The application should enable customers to browse products, search and filter items, manage their shopping carts, securely complete purchases, and track their orders.

As a React Native developer, you have been assigned to design, develop, test, and document the mobile application using React Native, JavaScript or TypeScript, and appropriate supporting libraries. The application must integrate with the provided API to retrieve product information, authenticate users, process orders, and manage customer transactions.

Tasks

1. **Develop product browsing screens:** Create React Native screens that allow users to browse products, search by product name, filter products by category or price, view product details, and switch between multiple languages.
2. **Implement user authentication:** Develop registration, login, logout, and password recovery features. Use appropriate authentication mechanisms to protect customer accounts and restrict access to authorized functions.
3. **Develop the shopping cart and checkout module:** Implement functionality that allows customers to add products to the cart, update quantities, remove items, calculate the total cost, and proceed to checkout.
4. **Implement order management:** Develop screens that display customers' order histories, order details, and order status. Integrate with the provided API to retrieve updated tracking information and display order progress.
5. **Integrate the provided API:** Connect the React Native application to the backend API using an appropriate HTTP client, such as Fetch API or Axios. Handle API responses, loading states, errors, and network connectivity problems.
6. **Conduct application testing:** Test the application to verify functional correctness, usability, performance, security, and compatibility across Android and iOS devices or emulators.
7. **Generate reports and documentation:** Provide appropriate reports, including product listings, customer account information, shopping cart summaries, order histories, sales summaries, testing results, and application performance reports, according to the data and reporting functions supported by the API.
8. **Implement navigation and state management:** Use an appropriate navigation library, such as React Navigation, to manage movement between screens. Apply a suitable state management

approach to maintain product, authentication, cart, and order information consistently throughout the application.

Additional Information

9. A backend API has been provided and must be integrated into the React Native application.
10. Login and authentication must be implemented as a separate module within the overall system, with appropriate navigation and access control.
11. The application must support both Android and iOS platforms.
12. The developer may use suitable React Native libraries for navigation, form validation, API communication, multilingual support, and state management.
13. The application must provide appropriate validation messages, error handling, and user feedback.
14. All developed features must be demonstrated, tested, and documented before submission.

Expected Deliverables

- React Native source code for the complete mobile application.
- A functional authentication module.
- Product browsing, search, and filtering screens.
- Shopping cart and checkout functionality.
- Order tracking and order history screens.
- Integration with the provided backend API.
- Testing report and screenshots of the implemented features.
- Technical documentation and a final project report.